KDB Fundamentals + Advanced
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Course Contents

1. The Basics
   - KDB+/Q Introduction
   - Salient Features
   - Variables / Assignments
   - Functional and Vector Programming
   - Time-series and Columnar Databases
   - Kdb+ startup options
   - Starting and exiting a session
2. Datatypes
   - Data Structure - Atom, List, Dictionary
   - Type numbers
   - Casting
3. List
   - Uniform and mixed lists
   - Indexing
   - Manipulation, Amendment
   - Various list operations (#, ?, sublist, cut, drop etc)
   - Generating random values
4. Dictionaries
5. Basic Operations
   - Order of evaluation
   - Arithmetic Operations
   - Apply - @ and .
6. Functions
   - Writing own function
   - Passing Parameters
   - Default parameters
   - Projection

- Global & local scope
- Use of @ and .
- Lambda function

7. Execution Control
   - Conditional Evaluation
   - Protected Execution
   - Signal
   - Loading Script
   - Passing Arguments & Parameters
   - Debugging & Error Messages
   - Internal Functions
8. Tables
   - Creating tables
   - Schema
   - Attributes
   - Virtual Columns (Date, I, x)
   - Sym File & Column
   - Enumeration
   - Table manipulations
   - (Select/Insert/Delete/Update)
9. Queries
   - Functional Form
   - Grouping & Aggregations
   - Use of fby, xbar
   - Query Performance & Optimization
   - Foreign Keys
   - Joins
10. I/O
    - Loading data to and from data file
    - Saving to file
    - Saving as CSV, XML, TXT

11. Types of DBs
    - Splayed
    - Partitioned
    - Segmented
    - On-Disk Tables
    - Saving Down Data on Disk (.Q.dpft, .Q.hdpf)
    - Use of attributes in on-disk and in-memory tables
    - Modifying schema of on-disk tables (HDB maintenance)
12. IPC
    - Message Handlers (.z.po, .z.pg, .z.ps, .z.pc)
    - Sync & Async calls
    - Callbacks
13. Timer function using .z.ts
14. System Commands & Variables
15. .Q & .z namespaces
16. Tickerplant Architecture
    - Feed Handler, TP, RDB, HDB, GW
    - TP Setup Standard Library (tp.q, u.q, r.q)
    - init, upd, sub, eod Functions
    - .u Namespace
17. Recovering RDB Through TP Log File